Reviewer and editor comments are in italics. Our replies are in plain font.

Associate Editor – comments:

TIM is a promising open-source tool that could make high-resolution imaging of tree cores and cookies more accessible. The manuscript is generally clear, and the combination of low-cost hardware, automated batch scanning, and on-device stitching is potentially useful. However, the evidence currently presented is not sufficient to evaluate image quality, operational performance, and readiness for adoption.

We thank the editor for their positive thoughts on TIM and were excited to hear that they and the reviewers saw promise in our new tool. We apologize that we failed to submit high-resolution images and other more detailed information with our initial submission and have worked to rectify these omissions, as detailed further below.

The most important point is that the central claim of very high resolution can't currently be judged. Because no full-resolution stitched images are provided and the figures are low-resolution thumbnails without scale bars. Please include representative full-resolution examples for both a core and a cookie. Please also distinguish pixel sampling from effective optical resolution and reconcile the inconsistent values in the text.

We agree that stitched images are necessary to evaluate the image quality. Thus, we have added full-resolution scans for multiple species in the supplementary material (Table S3). We reconciled the inconsistent values in submission abstract.

We now clarify that the resolution of the scan is based on the analog zoom adjustment of the lens body. Our implementation has the lens' zoom at about half of its capability leading to a resolution measurement of at least 21,000 DPI. We now describe this on line 311.

We now provide an estimate of the effective optical resolution using a 1 mm stage micrometer with 0.01 mm graduations. A cropped image of the stage micrometer taken by TIM is shown below in Figure 1 (this image and Fig. 2 are both now provided in the supplement).

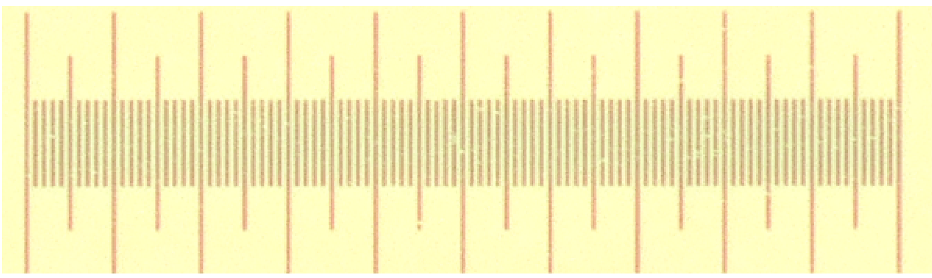

Figure 1: An image from TIM cropped to the extent of a 1 mm stage micrometer with 0.01 mm graduations.

When the image is cropped further to focus on four graduations (Figure 2), it is possible to count about 4 pixels between each graduation mark. This indicates that we are able to distinguish features that are less than 10 microns.

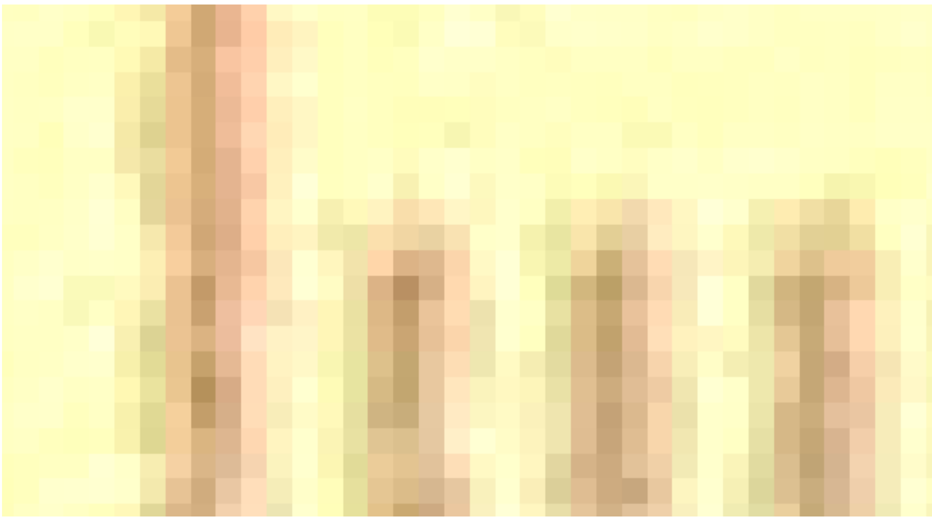

Figure 2: An image zoomed into four graduations of the aforementioned stage micrometer.

We believe pixel sampling allows us to set a good scale to measure between points on scans. It does not, however, inform the reader about how small of a physical feature it is able to distinguish in an image. We unfortunately do not own calibration sets to confirm the exact minimum (we did not budget for one), but we can buy one and run the tests if required.

Two practical points need clarification: whether autofocusing only every 5 to 10 frames affects sharpness or stitching on uneven samples, and the generality of the particle-smearing step.

This is a great point. As this is something we ourselves added and found to work, we can fully understand that others would be surprised as well. To address this we have provided extended text about this in the revised manuscript between line 201 and line 213, which we quote below in response to Reviewer #2.

To explain briefly here as well, autofocusing every 5 to 10 frames is the result of the sample maintaining a nearly constant distance from the camera sensor, but we test that the autofocus on each image appears sharp. While it is possible for TIM to autofocus every frame, we have found this approach works well and chosen to only refocus the lens more often than every 5-10 frames only when adjacent frames have significantly different focus scores.

We have also added specificity into the use case of the particle smearing method between line 269 and line 280.

Finally, please clarify how long and how intensively TIM has been in routine use, including approximate numbers of samples, species and operators, the period of use, and *any failure modes. This would reassure the resource-limited labs you are targeting. Please also address the suggestion both reviewers list, and add scale references to the sample images throughout.*

We appreciate the editor and reviewers highlighting the need to provide more information about our usage of TIM. We finished a working version of TIM in May 2025 and have used it near continuously since then. Over this time, we have processed approximately 1,300 cores, 80 cookies, and 20 species by 4 operators. This includes an intensive period where many cores were scanned in a three month period with batches of 25 cores were scanned one to three times a day (showing it works well even when heavily used). Batches were often left scanning throughout the night without supervision. Additionally, we have provided the design files to another lab and they have built their own TIM so it is now used across two labs (in two different countries).

We have updated the manuscript to outline this between line 306 and line 308.

These are all addressable, and I look forward to seeing the revised version.

We thank the editor for the hopeful comments and are excited to share the revised manuscript.

Reviewer # 1 – comments:

The authors present an interesting open-source contribution to dendrochronological imaging. The use of low- cost optics is genuinely innovative, and the sub-micron resolution potential could open new possibilities for wood anatomical analysis. However, several aspects require clarification to assess the potential of these low cost cameras.

We thank the reviewer for their positive comments and were happy to hear they see promise in the design. We appreciate that we should have provided more images and details in and with the manuscript, as we now do. We detail these changes further below.

The most pressing concern is image quality. The pixel size of the Raspberry Pi HQ sensor (1.55 μm × 1.55 μm) is substantially smaller than sensors used in prosumer cameras currently used by existing systems. While the authors claim high resolution, no final stitched images are provided for evaluation to assess image quality. The low-resolution thumbnails in Figures 4 and 5, which also lack scale bars, make it impossible to assess whether the resolution gain includes reasonable image quality. This is particularly important given that the key innovation of such an advance in resolution is enabling anatomical measurements (moving forward current 4 microns from existing platforms is not necessary for annual rings or earlywood/latewood). I encourage the authors to include representative final images at full resolution.

We thank the reviewer for highlighting that without representative full-resolution images, it is not possible to assess if the resolution gain improves the image quality appreciably. We have included multiple scans from different species in the supplementary material. Additional thumbnail images for the figures have been included to better showcase the resolution.

We added representative scans to the supplementary material (Table S3).

Regarding Figure 6, the authors would benefit from revising the axes so that time appears as a function of sample size, separating cores from cookies, and increasing the number of samples to allow for variability assessment. In its current form the figure is difficult to interpret and I cannot get a good idea of the time necessary to run a core (still the most common way of dendrochronological sampling). Does a typical core 350 mm with 5 mm width(1750 mm2) requires 150 minutes? I do not believe that if this were true you were presenting this article. So please, revise this table.

We thank the reviewer for their insight to improve the legibility of the figure. We have changed the figure so that time appears as a function of the sample's size, separated cores from cookies, and increased the number of samples.

The particle smearing approach is imaginative but raises some concern. In my experience with similar systems, additional surface treatment has never been necessary for successful stitching. The example shown in Figure 5 corresponds to a gymnosperm with highly distinctive tracheid structures, I am shocked that stitching has problems with such an easy sample. It would be informative to know how TIM performs with diffuse-porous angiosperms such as Betula or Populus, where informative tracheid pattern is absent and wood is very white. Additionally, silicon carbide is a very hard material. In TIM configuration the machine is over the samples, so it should not be a problem, but I would be worried if a lab with optical material close to TIM is smearing sand in a routine fashion. I have lived in the coast for many years and there is nothing so disturbing than the way sand gets into every crack.

We appreciate the reviewer's comment and understand their concerns. We agree that many samples will not need additional surface treatment to be successfully stitched. We have thus changed the description used in Figure 5 (main manuscript text) to specify the why this is a hard-to-scan sample which required additional surface treatment. Additionally, we have added specificity into the use case of the particle smearing method between line 269 and line 280. This includes highlighting that most samples should not need this approach.

Regarding autofocusing, the authors indicate that autofocusing occurs approximately every 5-10 frames. Existing systems typically autofocus every frame. The authors should clarify whether this trade-off affects final image sharpness, particularly in samples with surface irregularities and how it affects to sharpness and stitching.

We appreciate the concern of Reviewer 1 and have clarified how autofocusing every few frames affects the final image sharpness between line 201 and line 213. This clarification is quoted below.

> This approach of auto-focusing only when deviations from the previous sharpness are found ensures that all neighboring images have functionally the same sharpness, while minimizing the time spent on autofocusing.

> While scanning well prepared samples, we generally saw an autofocusing routine for every 5-10 images. All of the images captured with this method maintained high image sharpness throughout the sample and were well suited for stitching. This time saving feature can be tuned in the configuration file to have a more stringent or lenient sharpness threshold. The autofocusing procedure will run more frequently when the sample is not aligned well to the camera and in cases where the sample is less even. However, samples that have visible scoring marks from sanding, noticeable non-planar surfaces, or other major surface irregularities will likely have difficulties scanning with or without adaptive autofocusing. We have found that such samples simply require more attentive sanding to ensure a blemish free, planar surface for successful and rapid scanning.

A minor but important point concerns the scale of measurement. The authors invest considerably (given the cost of TIM) in a precision micrometric scale for calibration, yet no scale reference appears in any of the sample images presented. For a measurement tool, every sample should include a scale reference. A simple millimetric reference would suffice (you can improve accuracy by measuring more milimeters in the scale) and that would allow readers to verify the reported DPI values independently.

We agree with the reviewer that a scale reference on the scan is necessary to verify the DPI. To address this, we have provided scans with a few different scale references. These include a scan of a caliper with millimeter graduations, a cookie scan with a paper ruler on its surface, a photo of a 1 mm stage micrometer with 0.01 mm graduations, and digital scale bars on each core scan. We do not provide physical references on top of cores because of the high-resolution and precise focusing of TIM—the slight addition of a change in depth from a paper scale bar causes significant blurring in the images. Cookies have enough surface area for a millimetric reference to be placed on the surface while reserving a useful portion of the sample unaffected by the blurring caused by the physical scale.

The statement in line 300 indicating a maximum core length of 1500 mm at full resolution warrants clarification, as this exceeds the practical dimensions of any standard increment core and it is complicated to work with a "cookie" that size.

We thank the reviewer for pointing out that the maximum core length exceeds practical dimensions of increment borers. We have updated the text to clarify this on line 323.

A final point, more philosophical in nature but worth raising. There is an important difference between a tool that works and a tool that is ready for others to use. In my experience, that difference only becomes apparent after months of routine use in a real lab setting: different samples, different operators, unexpected failures. The question for the authors is simple: is TIM currently part of their day-to-day data collection, or was this manuscript written closer to the assembly stage? This matters because the dendrochronology community has seen promising methods published before they were truly mature, and the labs that tried to adopt them paid the price in time and frustration. TIM is explicitly aimed at labs with limited economic capacity, which makes this question even more important, those labs have less margin to troubleshoot. The authors should provide an account of how long and how intensively the tool has been *in use. This point is relevant to build confidence in the method.*

The reviewer raises valuable points when assessing the readiness for others to adopt TIM. While TIM does not have the polish of a commercial product, TIM has scanned over 1000 samples by at least 4 different operators over the past two years, as we now explain this in the text, between line 306 and line 308 as well as between line 332 and line 335.

Additionally, we shared our design files with another lab (based on another continent) and that lab has recreated TIM and used it for their research. While we continue to work to make TIM easier to build and use, we have done our best to provide labs who want to replicate the design with the resources they need to reproduce TIM. All of TIM's design is open-source and we are willing to assist with reasonable inquiries regarding the replication process.

In a nutshell, I love the approach, but I would be more convinced if the issues that I am presenting could be clarified.

We thank the reviewer for the positive comment and are happy to provide a revised manuscript to clarify their concerns.

Reviewer # 2 – comments:

This manuscript describes the utility for and design of an open-source system for scanning cores and cookies with an automatic stitching of high resolution images to a larger mosaic for efficient analysis. The justification is clearly made in the introduction. The description of the methodology is logically described and is mainly clear and easy to follow. In most cases where this reader formed a question, the next section inevitably answered the query.

We appreciate the reviewer's positive comments on the manuscript.

The system describes is very practical and I like that there are manual alternatives to the automated queuing approach. Likewise for cookies/disk material, specific radii can be manually selected to save time and file storage space/size. It would be useful to note what happens when a storage limit is hit -what does the system do in such a case? OK, I see that uncompressed array files are saved – but does this allow stitching?

We thank the reviewer for highlighting the need for clarification regarding file size limits. When the file size limit is reached, the system still stitches the sample in the same method but it is saved as a memory mapped NumPy array instead of a tiff image. The manuscript has been updated with this clarification on line 326.

Its great that the study even tests the use of a grain enhancer like silicon carbide. I presume that more dramatic stains would not affect imaging and stitching contrast settings?

More dramatic stains would not affect the imaging and stitching contrast settings. We chose to use silicon carbide as we found simple staining techniques did not resolve the stitching issue.

My last question is also answered about the potential for ring or feature detection. I hope that is something your team considers as it seems like the next logical step – and would then provide serious competition to the WinDendro systems. Bravo!

We thank the reviewer for their enthusiasm about the potential for ring and feature detection. We are hopeful that future methods for ring and other feature identification are developed and can be integrated with TIM through its open-source design.

Abstract point 3: It's a little unclear with the current notation what "21 140 DPI" means – should it be 21,140 DPI? Also, one version of your abstract indicates 0.83 pixels per micron and the other 1.2 microns per pixel. Similarly for Point 3, is the value $3,000 USD? The problem seems to be a space has been introduced to replace the thousand separators: L54 – 6 400 DPI rather than 6,400 DPI. See L67, L69, L88, L89... etc. throughout the manuscript.

We changed the notation to use commas in our number notation. Additionally the abstract was revised to only display microns per pixel.

L96: Change "current" to currently

Done. See line 96.

L129: Should read "...and have a planar surface for..."

Done. See line 131.

L137: Small point admittedly, but conventionally the reference to a figure should appear before the figure itself occurs. Same for Figure 3 which appears before the reference to it.

Done. Moved figure reference above figure.

L160: Gu et al. should not be within parenthesis given the current structure, only the publication date (2015).

Done. See line 162.

L185: Clarification needed. "...same threshold was used across all cores for 5 species..." Is the point that the sequence worked for variation caused by having different species, or that the transition from one species to another was handled?

Done. See line 186.

Section 2.4.2: This description needs a little rephrasing for better clarity. Quite a number of sentences repeat words (code was generated to convert code, package similarly. Again final sentence -code was generated for codebase integration and was noted *in code... etc. I think I understand in general what you mean, but I'm not sure. Please rephrase more carefully.*

We thank the reviewer for addressing the repetition of words in the manuscript. We have rephrased the section to improve clarity between line 297 and line 304.

Figure S1: The scales in the two images are very difficult to read for the ordinary human, slightly larger font would be a help. The scale in the (b) image currently seems to be out of sync with the image.

We have updated the figure by cropping the image to the extent of the stage micrometer and removing the scales as they did not add useful information to the figure.

So with a little revision I suggest that this will be a really interesting paper and provide an excellent tool for a section of cross disciplinary research that is crying out for such a resource. Well done.

We appreciate the positive comments and the acknowledgement of the potential of our tool. We hope other groups will build and use TIM or develop further on its design.